

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df=pd.read_csv(r"C:\Users\Janvi Kapse\Downloads\student_performance_100_rows.csv")
print(df)
```

	Student_ID	Study_Hours	Attendance	Exam_Score
0	S001	1.5	68	42
1	S002	2.0	72	46
2	S003	2.5	75	50
3	S004	3.0	78	55
4	S005	3.5	80	58
..	...	...	...	...
95	S096	4.6	84	68
96	S097	5.2	87	72
97	S098	5.8	90	77
98	S099	6.5	92	83
99	S100	7.0	95	88

[100 rows x 4 columns]

```
In [2]: df.head()
```

```
Out[2]:
```

	Student_ID	Study_Hours	Attendance	Exam_Score
0	S001	1.5	68	42
1	S002	2.0	72	46
2	S003	2.5	75	50
3	S004	3.0	78	55
4	S005	3.5	80	58

```
In [3]: df.tail()
```

```
Out[3]:
```

	Student_ID	Study_Hours	Attendance	Exam_Score
95	S096	4.6	84	68
96	S097	5.2	87	72
97	S098	5.8	90	77
98	S099	6.5	92	83
99	S100	7.0	95	88

```
In [7]: df.describe()
```

Out[7]:

	Study_Hours	Attendance	Exam_Score
count	100.000000	100.000000	100.000000
mean	4.231000	81.710000	64.320000
std	1.518152	7.659806	12.295092
min	1.500000	64.000000	40.000000
25%	2.975000	76.000000	54.750000
50%	4.250000	83.000000	65.000000
75%	5.500000	88.000000	74.250000
max	7.000000	95.000000	88.000000

In [6]: `df.shape`

Out[6]: (100, 4)

In [8]: `df.info()`

```
<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Student_ID  100 non-null    str
1   Study_Hours 100 non-null    float64
2   Attendance   100 non-null    int64
3   Exam_Score   100 non-null    int64
dtypes: float64(1), int64(2), str(1)
memory usage: 3.3 KB
```

In [9]: `df.columns`

Out[9]: Index(['Student_ID', 'Study_Hours', 'Attendance', 'Exam_Score'], dtype='str')

In [10]: `df.isnull().sum()`

Out[10]: Student_ID 0
Study_Hours 0
Attendance 0
Exam_Score 0
dtype: int64

In [15]: *# Calculate correlation between Study Hours and Exam Score*
`correlation = df['Study_Hours'].corr(df['Exam_Score'])`

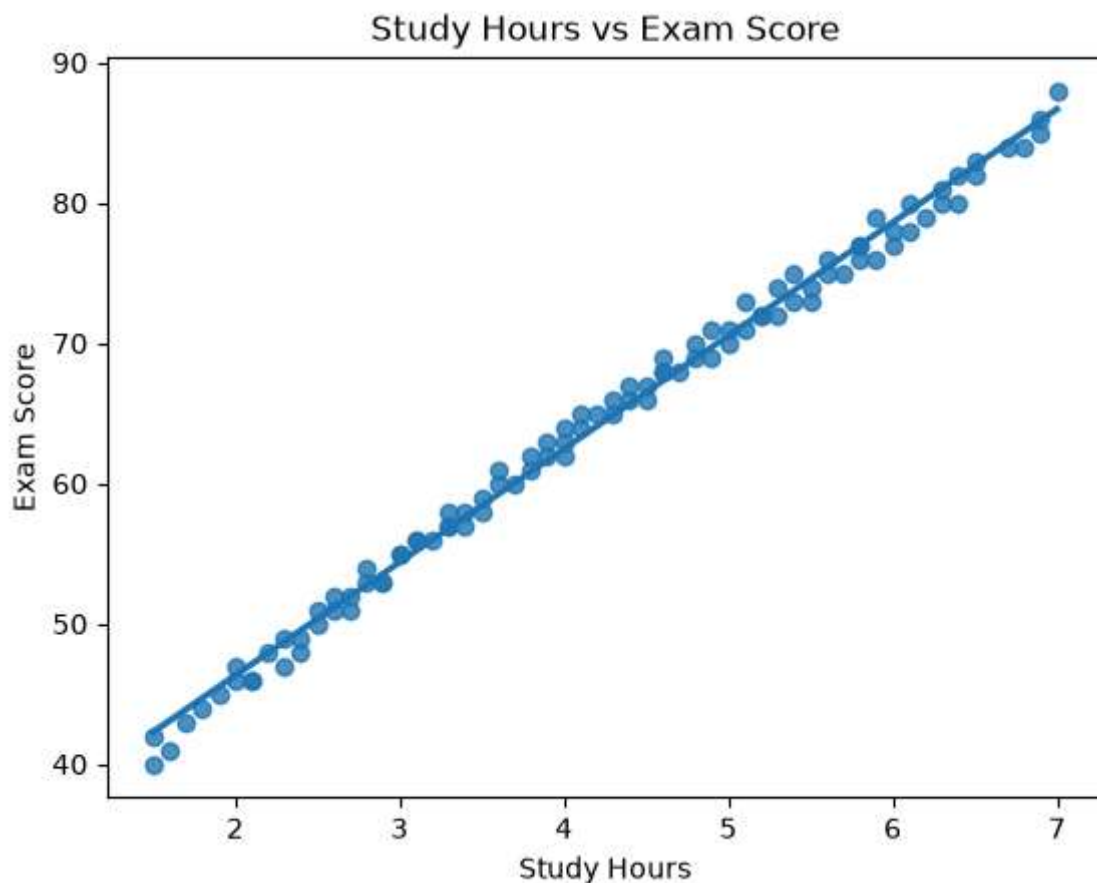
Display correlation
`print("Correlation:", correlation)`

Correlation: 0.9970731268039084

```
In [16]: # Create scatter plot with regression line
sns.regplot(x='Study_Hours', y='Exam_Score', data=df)

# Add title and Labels
plt.title("Study Hours vs Exam Score")
plt.xlabel("Study Hours")
plt.ylabel("Exam Score")

# Display plot
plt.show()
```



```
In [17]: # Select required columns
correlation_matrix = df[['Study_Hours', 'Exam_Score']].corr()

# Display correlation matrix
print(correlation_matrix)
```

	Study_Hours	Exam_Score
Study_Hours	1.000000	0.997073
Exam_Score	0.997073	1.000000

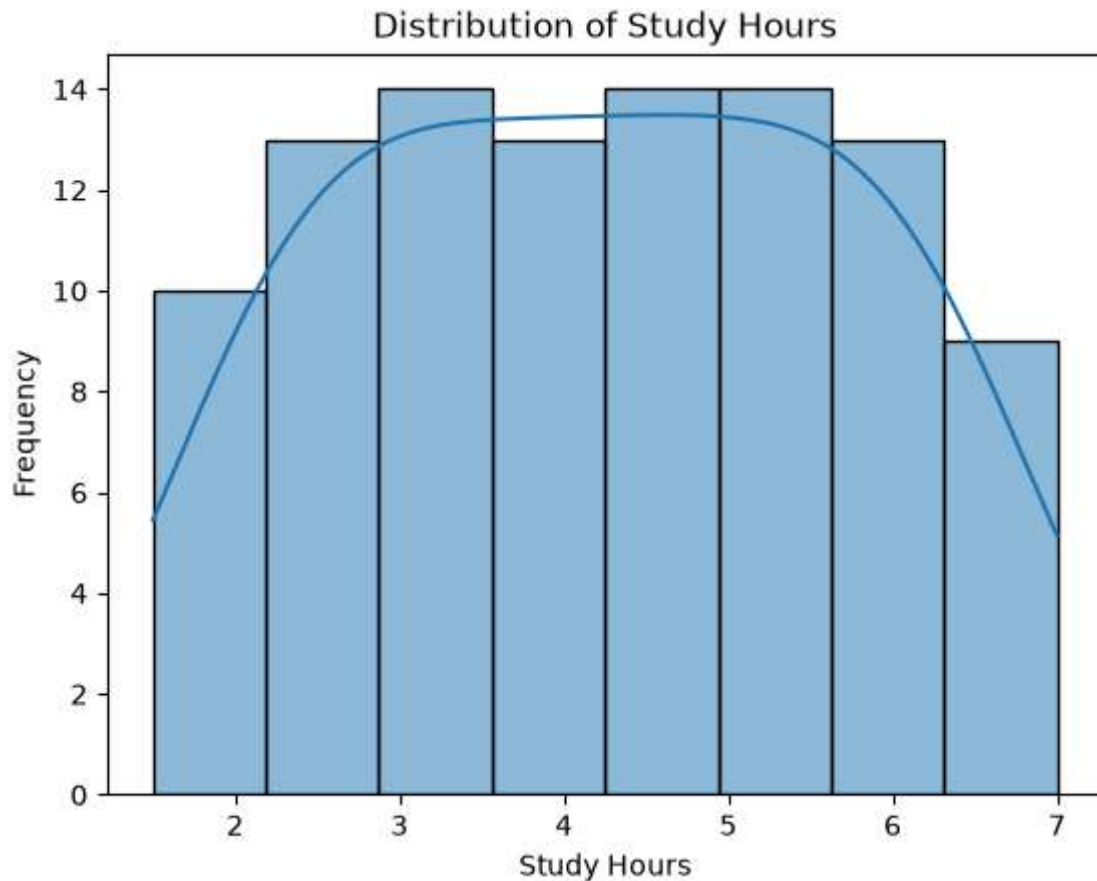
```
In [19]: # Calculate skewness of Study Hours
study_hours_skewness = df['Study_Hours'].skew()
# Calculate skewness of Exam Score
exam_score_skewness = df['Exam_Score'].skew()
print("Study Hours Skewness:", study_hours_skewness)
print("Exam Score Skewness:", exam_score_skewness)
```

Study Hours Skewness: 0.00035304014195932705

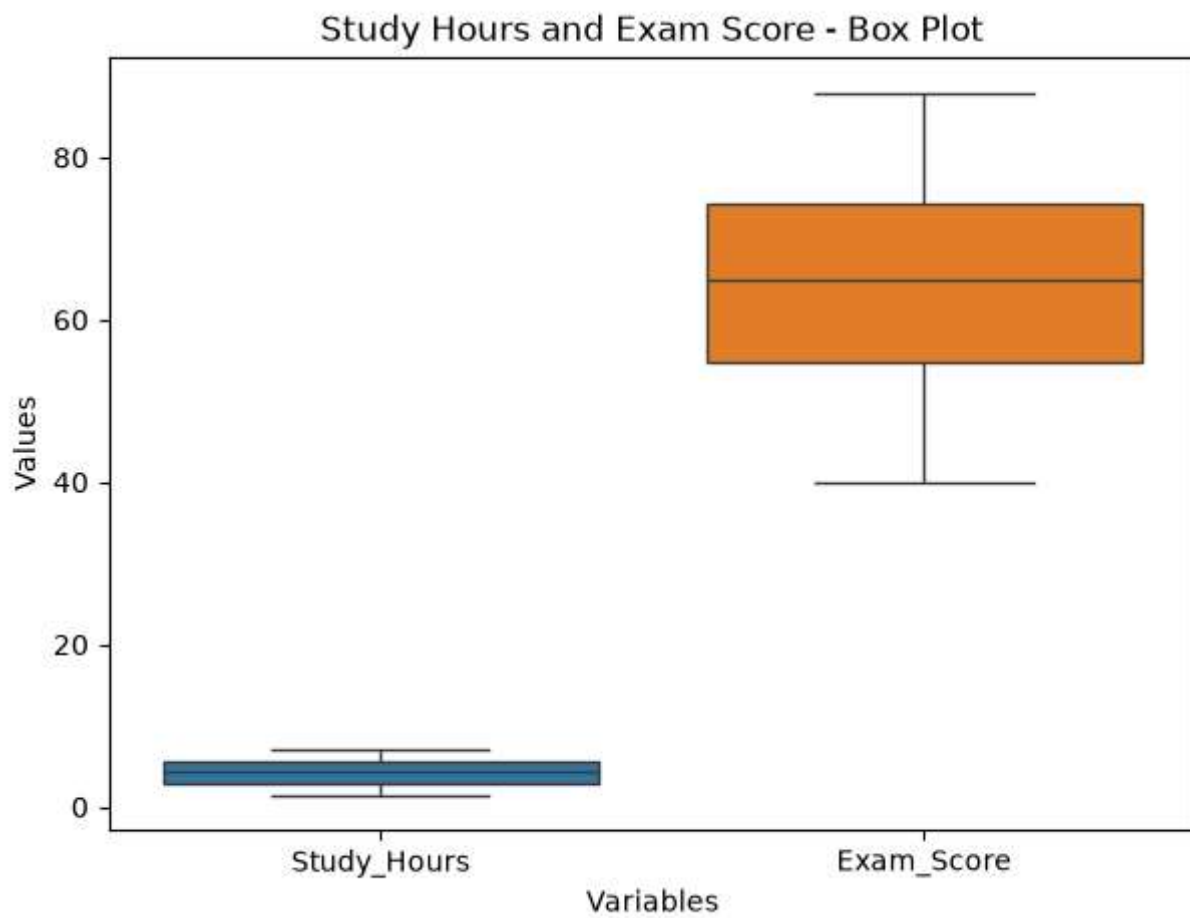
Exam Score Skewness: -0.10473520515950736

```
In [20]: # Plot histogram and KDE for Study Hours
sns.histplot(df['Study_Hours'], kde=True)
plt.title("Distribution of Study Hours")
plt.xlabel("Study Hours")
plt.ylabel("Frequency")

plt.show()
```



```
In [23]: # Create box plots of both variables
plt.figure(figsize=(7,5))
sns.boxplot(data=df[['Study_Hours', 'Exam_Score']])
plt.title("Study Hours and Exam Score - Box Plot")
plt.xlabel("Variables")
plt.ylabel("Values")
plt.show()
```



In []: